



KEMENTERIAN PENDIDIKAN TINGGI
JABATAN PENDIDIKAN POLITEKNIK DAN KOLEJ KOMUNITI



INFORMATION & COMMUNICATION TECHNOLOGY DEPARTMENT SESSION 1 2026/2027		
DFK30053 OBJECT ORIENTED PROGRAMMING		
LECTURER NAME		
TYPE OF ASSESSMENT	MINI PROJECT	
TOPIC	T1 : OBJECT-ORIENTED PROGRAMMING AND JAVA BASICS T2 : CLASSES, METHODS, AND INPUT/OUTPUT STREAMS IN JAVA T3 : NORMALIZATION AND ENTITY RELATIONSHIP DIAGRAM T4 : STRUCTURED QUERY LANGUAGE (SQL) T5 : PACKAGES AND EXCEPTION HANDLING	
DURATION	2 HOURS (F2F), 6 HOURS (NF2F)	
DATE OF ASSESSMENT		
STUDENT'S	NAME	REGISTRATION NO.
INFORMATION		
TOTAL MARKS	CLO 2	/ 100 MARKS

DFK30053 OBJECT ORIENTED PROGRAMMING			
MINI PROJECT			
CLO2	Build Java programs using fundamental Object-Oriented Programming (OOP) concepts and basic data structures.	PLO3	DIT

INSTRUCTION: Choose ONE (1) of the following questions.

QUESTION 1

Scenario: A group of college students plans to organize a Campus Gaming Tournament involving popular competitive games. The organizer needs a simple Java-based system to manage player registration, game categories, tournament participation and match information.

Currently, player and tournament information is managed manually through chat messages and spreadsheets. This makes it difficult to search player information, manage different types of players, calculate tournament fees and display tournament records.

You and your team are required to design and develop a Java-based Gaming Tournament Management System using Object-Oriented Programming concepts.

The system should allow the organizer to register players, manage game information, create tournament registrations, calculate registration fees and display tournament information.

The system must demonstrate the following concepts:

- Object-Oriented Programming
- Classes and methods
- Input/Output
- Control structures
- Arrays / ArrayList
- Inheritance
- Polymorphism
- String manipulation
- Packages
- Exception handling

As additional information, the system should have:

1. Main menu

- The system should provide a menu similar to:

```
=====
CAMPUS GAMING TOURNAMENT SYSTEM
=====
1. Add Game
2. Display Games
3. Register Player
4. Create Tournament Registration
5. Display Tournament Registration
6. Calculate Registration Fee
7. Exit
=====
Enter your choice:
```

The program should continue running until the user selects **Exit**.

2. Game Management

- The system must allow the organizer to add and display games.
- Each game should contain:
 - a. Game ID
 - b. Game Name
 - c. Game Category
 - d. Registration Fee
 - e. Maximum Players

3. Player Management

- The system must allow the organizer to register players.
- Each player should contain:
 - a. Player ID
 - b. Player Name
 - c. Username / Gamer Tag
 - d. Phone Number
 - e. Age
- The system should be able to display all registered players.

**** Any additional information is based on group creativity.**

QUESTION 2

Scenario: A local event company plans to organize several music events and concerts featuring different artists and music genres. The organizer needs a simple computerized system to manage concert information, artist information, customer bookings and ticket sales.

Currently, the organizer manages event and ticket information manually using spreadsheets and messaging applications. This makes it difficult to keep track of available seats, ticket categories, customer information and total ticket charges.

You are required to design and develop a Java-based Music Event & Concert Ticketing System using Object-Oriented Programming concepts.

The system should allow the organizer to manage music events, register artists, record customer information, sell tickets, calculate ticket charges and display booking information.

The system must demonstrate the following concepts:

- Object-Oriented Programming
- Classes and methods
- Input/Output
- Control structures
- Arrays / ArrayList
- Inheritance
- Polymorphism
- String manipulation
- Packages
- Exception handling

As additional information, the system should have:

1. Main menu

- The system should provide a menu similar to:

```
=====
MUSIC EVENT & CONCERT TICKETING SYSTEM
=====
1. Add Music Event
2. Display Music Events
3. Register Artist
4. Register Customer
5. Book Concert Ticket
6. Display Ticket Booking
7. Exit
=====
Enter your choice:
```

The program should continue running until the user selects **Exit**.

2. Music Event Management

- The system must allow the organizer to add and display music events.
- Each event should contain at least:
 - a. Event ID
 - b. Event Name
 - c. Music Genre
 - d. Event Date
 - e. Venue
 - f. Ticket Price
 - g. Available Tickets

3. Artist Management

- The system must allow the organizer to register artists who will perform at the event.
- Each artist should contain:
 - a. Artist ID
 - b. Artist Name
 - c. Stage Name
 - d. Music Genre
- The system should be able to display all registered artists.

4. Customer Management

- The system must allow customers to register before purchasing tickets.
- Each customer should contain:
 - a. Customer ID
 - b. Customer Name
 - c. Email
 - d. Phone Number

5. Ticket Booking

- The system must allow customers to book tickets for a selected music event.
- The booking should record information such as:
 - a. Booking ID
 - b. Customer ID
 - c. Event ID
 - d. Ticket Type
 - e. Quantity
- The system should update the number of available tickets after a successful booking.

**** Any additional information is based on group creativity.**

PREPARED BY:  ----- (ZALINDA BT IBRAHIM) PEGAWAI PENDIDIKAN PENGAJIAN TINGGI JABATAN TEKNOLOGI MAKLUMAT DAN KOMUNIKASI POLITEKNIK UNGKU OMAR, IPOH PERAK DARUL RIDZUAN DATE: 23/9/2026	CHECKED BY:  ----- (MOHD HAZRI B. HASNAN) Pegawai Pendidikan Pengajian Tinggi Jabatan Teknologi Maklumat Dan Komunikasi Politeknik Ungku Omar 31400 Ipoh, Perak DATE: 23/9/2026	APPROVED BY:  ----- (SHARIFAH NUR BINTI SYED ISMAIL) Ketua Program Diploma Teknologi Maklumat (Trek Sistem Rangkaian) Jabatan Teknologi Maklumat & Komunikasi Politeknik Ungku Omar, Ipoh Perak. DATE: 23/9/2026
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Rubric:

**PLO3 – Perform support and development tasks on Information Technology solutions related to job functions.
CLUSTER 3A – CLS3A: Practical Skills (P4)**

Criteria	Performance Level				Weightage	Standard	Score
	Excellent 4	Good 3	Average 2	Poor 1			
Practical Skills							
1. Class & Object Implementation	Implements appropriate classes, attributes, constructors and methods accurately; objects are created and used effectively according to the system requirements.	Implements most required classes, attributes, constructors and methods correctly with minor errors.	Implements basic classes and objects but some attributes or methods are incomplete or incorrectly implemented.	Classes and objects are largely incomplete or incorrectly implemented.	15%	[1/4*15	
2. Inheritance & Polymorphism	Correctly implements inheritance and method overriding; polymorphism is clearly applied in the working system.	Implements inheritance and polymorphism correctly with minor implementation errors.	Implements inheritance but polymorphism is limited or partially functional.	Inheritance and polymorphism are not appropriately implemented.	15%	[1/4*15	
3. Program Functionality & Control Structures	Develops a fully functional program; selection and repetition structures are appropriately integrated to perform all required operations.	Most required functions work correctly with minor errors in control structures.	Basic functions work but several operations or control structures require correction.	Major functions are incomplete or do not work correctly.	20%	[1/4*20	
4. Data Structures & Data Processing	Effectively uses arrays/ArrayList to store, retrieve, search and process multiple records accurately.	Uses appropriate data structures with minor errors in data processing.	Uses a basic data structure but implementation or processing is partially functional.	Data structures are inappropriate, incomplete or not functional.	10%	[1/4*10	

Course Policy Statements

- Assessment items submitted more than five working days after the due date are awarded zero marks. Course lecturer(s) may use their discretion in handling any unusual cases of late submission/ absence for assessment activities on a case-by case basis, while adhering to the Head of Program.
- Politeknik Ungku Omar values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism, and other academic offenses. Any plagiarism will be awarded zero marks.

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